

Lalitha Sri Kakarala

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EDUCATION

- **University Of Central Missouri** Missouri, USA
Master of Science in Computer Science; GPA: 3.5/4
Aug 2023 – May 2025
Courses: Operating System, Cloud Computing, Design and Analysis of Algorithms, Database Management System.
- **Ramachandra College Of Engineering** Eluru, India
Bachelor of Technology in Computer Science and Engineering; GPA: 7.87/10.0
June 2018 – June 2022
Courses: Data Structures and Algorithms, Advanced Java Programming, Machine Learning.

SKILLS

- **Languages:** Java, Python, C, CSharp, HTML, CSS, JavaScript.
- **Cloud Services:** Amazon Web Services(Lambda, S3, VPC, EC2, API Gateway), Google Cloud Platform(Kubernetes, IAM, SQL).
- **Frameworks & Web Services:** Spring Boot, Django, Angular, Bootstrap, ReactJS, NPM, jQuery, REST, SOAP.
- **Tools:** NodeJS, IntelliJ, Visual Studio, Jupyter, Git, BitBucket, Docker, Maven, Jenkins, Helm, JWT, JUnit, Mockito
- **Containerization:** Docker, Kubernetes
- **Database Tools:** Microsoft SQLServer, MySQL, PostgreSQL, MongoDB, Oracle.
- **Machine learning:** Classification and Regression, Scikit-learn, Normalization, Standardization, NLTK, Basic Deep Learning, Distributed ML Systems.

EXPERIENCE

- **Adventa Soft Inc** Texas, USA
Intern Software Engineer
Aug 2024 - June 2025
 - Using HTML5, CSS3, JavaScript, jQuery, Angular, and Bootstrap, a comprehensive healthcare analytics platform was developed with an emphasis on an intuitive front-end interface, offering academics and doctors adaptable data visualization capabilities.
 - Using RESTful web services and modular design, I effectively managed over 10 TB of healthcare data by architecting the back-end using Node.js and Express.
 - Used Kubernetes to orchestrate cloud-native deployments and MongoDB to manage data storage, guaranteeing scalable and robust application infrastructure.
 - S3, EC2, and API Gateway AWS services were integrated to create all-inclusive cloud data processing and storage solutions.
 - By identifying important patterns in healthcare data, logistic regression and decision tree algorithms were used to improve treatment outcomes by 25%.
 - Used natural language processing (NLP) techniques to extract important insights from unstructured healthcare data, preprocessing big datasets and improving the platform's analysis capabilities by 45%.
 - Developed and maintained microservices that ensured modular, scalable backend architecture, aligning with healthcare compliance standards such as HIPAA.
 - Deployed containerized applications in Kubernetes clusters using Helm charts for consistent configuration management and simplified CI/CD.
- **Tata Consultancy Services** Hyderabad, India
Software Development Engineer
July 2022 - July 2023
 - Created a cloud-based deployment platform to streamline resource allocation and task scheduling across dispersed systems for Electronic Design Automation (EDA) workloads. used Lambda functions and AWS EC2 instances to manage task processing at scale while maximizing fault tolerance and speed.
 - Oversaw a cost-optimization project that found inefficiencies in resource allocation by examining LSF log data for EDA job scheduling. Through the development and implementation of Python machine learning models that forecasted resource requirements, this project achieved a 20% decrease in AWS expenses and earned praise from the client.
 - In line with Amazon's focus on creating scalable services technology, I designed scalable backend services using Spring Boot and Java, creating RESTful APIs to handle high-throughput queries in the application layer.
 - For data-intensive applications, optimized indexing and distributed storage solutions were designed and implemented. Low-latency queries, high fault tolerance, and low overhead costs were guaranteed by the implementation of a data retrieval layer.
 - Jenkins was used to implement CI/CD pipelines for automated builds, testing, and deployment, enabling continuous integration and speeding up iterations—two crucial components of the agile development environment that Amazon values.
 - Leveraged AWS CloudWatch and DataDog to monitor EDA job metrics and proactively alert anomalies, improving system reliability by 40%.
 - Refactored legacy components of the backend system using design patterns (Factory, Singleton), improving code maintainability and scalability.
 - Used Pandas and Scikit-learn for feature engineering and hyperparameter tuning, increasing model accuracy for cloud resource prediction by 18%.
- **V9 CARINAE** Hyderabad, India
Junior Software Engineer
July 2021 - Jun 2022
 - In charge of the development of NEURED, an EDA task execution platform that boosted job execution speed by 30%. On AWS, it was on show at re:Invent 2022.
 - Using Spring Boot, I created microservices and Rest APIs for an Electronic Design Automation Jobs platform, demonstrating my versatility and proficiency with contemporary programming frameworks.

- Using ReactJS, we created interactive and responsive web apps that increased user engagement by 30%.
- Lombok annotations and Spring Boot configurations were used to streamline JSON data processing and configuration administration.
- Kubernetes was used for automated deployment and scalability, and defined SOAP and RESTful API endpoints were consumed in React services.
- A 99% decrease in unwanted access attempts was achieved by implementing Spring Security. Additionally, NAT gateways, VPCs, and subnets were developed, apps were deployed on EC2, and S3 was used for storage.
- Developed effective and responsive applications that dynamically load data in response to user activities by utilizing Python for data analysis and back-end logic.
- The analytical team was able to access and analyze data 20% faster because to the simplification of data gathering across all apps inside the company and the aggregation of all repositories' varied testing criteria.

PROJECTS

• SmartCommute Tracker

Mar 2024 - May 2024

- Led the creation of a transit platform with Angular and Java and Spring Boot, enhancing functionalities including reservation management, ticket booking, and user authentication. This led to a 30% increase in operational efficiency and a 40% increase in customer engagement.
- Using tools like Spring Security, JWT, MySQL, and Maven, I ensured strong security and efficient development processes.
- Designed and documented RESTful APIs, improving clarity and usability for developers across teams.
- JUnit and Mockito were used to implement automated testing, which greatly decreased production bugs and increased system reliability by 25%.
- Ensured maintainable and effective software delivery by implementing best practices in code organization and version control with Git, which increased system scalability by 35%.

• Scalable Social Media Web Application

Aug 2023 - Nov 2023

- Using React.js and React Hooks, I oversaw the design and construction of a dynamic social media web application. developed a responsive, scalable, and safe platform using Google Firebase.
- Designed with HTML5, CSS, JavaScript, ReactJS Hooks, Router, and Bootstrap to provide a contemporary, captivating user experience. Redux and Firebase's database were used to integrate real-time messaging and networking capabilities.
- Simplified scalability and uptime by deploying and maintaining the application on AWS services.
- Cloudinary, an image optimization tool, was used to optimize photo uploads and downloads for effective image management, speed, and load times.
- By optimizing the platform for low-latency performance, which allowed for lag-free, fluid interactions, great user satisfaction was attained. With real-time communication features, the program caters to an expanding user base while maintaining a seamless experience even when under a lot of load.

• Real-Time Examination Management System

Oct 2020-Dec 2020

- In order to notify customers of technical problems and connectivity interruptions during tests, real-time notifications were integrated using web technologies, Node.js, AWS S3, Docker, Spring Boot, and PostgreSQL. 99% notification delivery was attained, guaranteeing consumers received timely alerts and enabling a seamless exam-taking experience.
- Exam pauses due to internet connectivity problems were reduced by 50% as a result of the alerts function, which enabled users to resume their tests without any problems.
- Improved the exam platform's UI/UX by using user feedback and interview data. User satisfaction ratings improved by 25% and user engagement increased by 30% as a result of these improvements. This beneficial effect on user psychology might have helped students do better on tests.